

Open-Set Object Retrieval in Clutter via Hybrid Vision-Language and Geometric Reasoning

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Abstract. Robots must retrieve novel objects from clutter, where the target may be neither visible nor directly reachable from a constrained viewpoint. This work presents a complete, modular pipeline for such targeted object retrieval in clutter and focuses on a critical component, i.e., deciding the order in which blocking objects must be removed before the target becomes available. Vision-Language Models (VLMs) have been proposed as pretrained solutions for spatial reasoning over RGB images, while engineered solutions heuristically infer object relationships from depth and grasp feasibility. Physics-based simulation on a new dataset of heavily occluded scenes and real-robot experiments show that a robotics-pretrained VLM used without fine-tuning underperforms random selection among graspable objects (68% vs. 79% success), while an engineered dependency graph does better (81%) but has its own failure modes. Hybrid strategies that combine the VLM’s visual reasoning with an engineered dependency graph improve performance: describing grasp-blocking relations to the VLM in language raises success to 87%, approaching the 94% of human experts, and retrieves the target in all 9 real-robot trials (3 scenes) without dropping any object. On scenes with objects in natural resting poses this hybrid matches the experts (94% vs. 90%); the remaining gap lies in unstructured piles (81% vs. 97%).

Keywords: Vision-language model, Robot manipulation, Task planning, Dependency graph, Mechanical search

1 Introduction

Targeted Object Retrieval in Clutter (TORC) arises in manufacturing, logistics, and service robotics, and brings together multiple challenges: perception, reconstruction of unknown object shapes, grasping, motion and task planning, i.e., identifying the sequence of objects to remove so as to retrieve a target.

In some setups, TORC can be engineered away by assuming known object models or homogeneous geometry, or by structuring the workspace to reduce clutter and guarantee visibility, at the expense of space efficiency and automation cost. This work instead targets an open set of unknown objects in the less structured, heavily occluded setups of Fig. 1: a single front view of a cluttered



Fig. 1. (Left, 2×2) Bird’s-eye view of a real-world TORC instance (top left) and the segmented robot-view images captured before each pick until the target (the dark box labeled 0, bottom right) is exposed and retrieved. (Right) The real-world setup: a robotic arm with a parallel gripper picks an object using a torso-mounted stereo camera for perception.

scene hides the target, and blocking objects must be removed until it is revealed and reachable. Clutter and kinematic constraints restrict the reachable grasps and safe retraction paths, especially when object models are unavailable and shape completion is unreliable. Moreover, when objects are tightly packed or stacked, removing one may cause others to fall off their support surface.

To address such TORC instances, this work first builds a complete, modular pipeline that brings together state-of-the-art tools for segmentation, grasp generation, and motion planning, and enables the isolated evaluation of a key task-planning component: selecting which occluding object to remove next. Engineered approaches construct scene or dependency graphs (DGs) [38, 19] from depth images to encode reachability and visibility relations such as “behind” or “below”. These heuristics become fragile as clutter and geometric complexity increase. Vision-Language Models (VLMs) have recently been proposed [28, 29, 2] as pretrained, model-free alternatives that provide spatial reasoning over RGB images. DG-based and VLM-based planners thus operate over different information—depth and grasp feasibility versus RGB images—which motivates hybrid planners that combine both.

This work provides a comparative analysis of TORC task planners in simulation and on a real robot. It **contributes**:

1. A complete, modular pipeline for TORC with unknown objects, significant clutter, a single viewpoint, and limited reachability, extendable to other robots, planners, and perception modules.
2. A dataset of 40 heavily cluttered and occluded simulated TORC problems, built on a grasping benchmark [1] and curated so that object selection critically affects retrieval success.
3. Two hybrid task planners for TORC object selection that integrate a VLM’s spatial reasoning with geometrically derived object relationships; they either let the VLM refine a dependency graph or describe grasp dependencies to the VLM.

4. Evaluation of VLM-, DG-, and hybrid task planners against random selection and an “expert” human-in-the-loop baseline, in 2,000 simulated and 45 real trials.

The evaluation reveals that a robotics-pretrained VLM used without fine-tuning does not yet possess sufficient spatial reasoning over images alone: it retrieves the target less often than random selection among graspable objects. Providing the VLM with geometric information from depth and grasp feasibility, however, yields the most reliable action selection for TORC, while reducing dependence on engineered heuristics. Code, prompts, the scene dataset, and supplementary material are available online.³

2 Related Work

Mechanical Search. When a target is occluded or inaccessible, an effective strategy is mechanical search [3], which sequentially rearranges objects to expose and retrieve the target. Early work exploits the visibility and reachability structure between objects in simple scenes with known geometry to formulate a rearrangement plan [4]. Belief-space search with a Gaussian process [23] aims to make all objects visible, which is not action-optimal. Reinforcement learning has improved action efficiency [22], albeit only with cuboids. Learned occupancy distributions (X-rays) locate the target without explicit relationship reasoning [9]; follow-up work added a combined suction-grasping tool [8] and stacked objects [10], under rigid assumptions on object geometry.

Scene/Dependency Graphs for Manipulation. Scene graphs have been used to plan picking for retrieval via a Regression Planning Network [38] or a POMDP [12], with similar formulations for manipulation and target discovery [37, 33]. Many approaches learn to detect object relationships that inform scene graph construction [18, 14, 21]. Others build robot-oriented traversability graphs of reachable object poses [20], and recent work combines object relationships with reachability estimates into a dependency graph over the manipulation order, yielding a resolution-complete retrieval algorithm [19]. Most task and motion planning works assume known models and rely on object pose estimation [32, 17], or address pose uncertainty at the motion-planning level [30].

VLM Planning. Prior efforts have used VLMs, directly or with fine-tuning, for spatial reasoning and object relationships instead of constructing dependency graphs explicitly [16, 35, 25]. Some have used VLMs for the entire retrieval pipeline (perception, grasping, and selection) [29] or alongside classical task and motion planning for longer-horizon tasks [13, 34]. More recent techniques build dependency or obstruction graphs with privileged information over large scene datasets to construct an expert policy for fine-tuning a VLM [5, 11]. Benchmarks for fetching in clutter [7] evaluate grasp-and-retrieve pipelines but not object selection under heavy occlusion. In contrast, our hybrids use a VLM without fine-tuning, either to guide dependency-graph generation or to interpret runtime reachability dependencies.

³ <https://2026vlmtaskplanningclutter.github.io/>

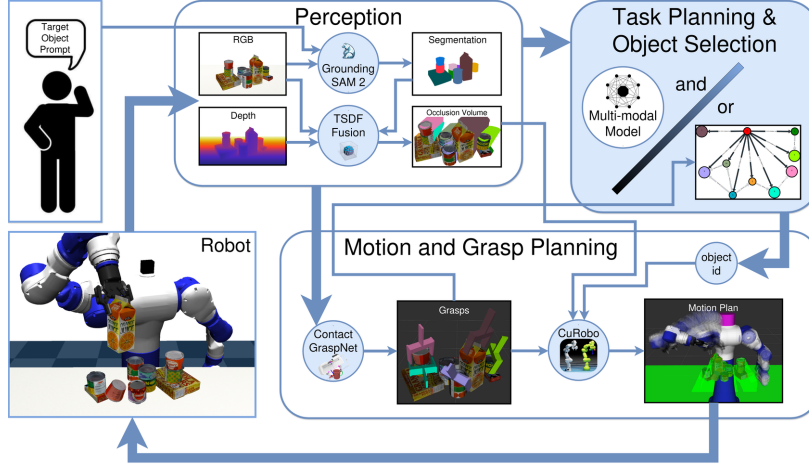


Fig. 2. The modular system architecture for retrieving occluded objects in clutter: perception (top middle), task planning (top right—the focus of this paper’s evaluation), and grasp and motion planning (bottom right).

3 TORC Problem Setup

Workspace. The workspace is a tabletop next to a robot arm with a pinch gripper and an RGB-D camera. It contains n unknown, heterogeneous objects $\mathcal{O} = \{o_1, \dots, o_n\}$ resting on the support surface or on other objects, which may not be directly graspable from any orientation. A specific object $o_t \in \mathcal{O}$, designated through natural language as the target, may be partially or completely occluded from the camera; the target and most other objects are expected to become graspable once exposed. Let $\mathcal{P}_k \subseteq \mathcal{O}$ be the objects picked and removed after the k^{th} action, $\mathcal{D}_k \subseteq \mathcal{O}$ the objects that have unintentionally dropped from the tabletop, and $\mathcal{R}_k = \mathcal{O} \setminus (\mathcal{P}_k \cup \mathcal{D}_k)$ the objects remaining at step k .

Input/Output. The robot’s input includes its proprioception, a natural language description of the target, and egocentric images $Z_k = (I_{RGB}(k), I_D(k))$ from the single, calibrated RGB-D camera viewing the front of the tabletop; no prior 3D models or poses are provided for any object. Each action $a_k = (o_i, g_j)$ selects an object $o_i \in \mathcal{R}_k$ and outputs a motion plan to retrieve it with grasp g_j .

Objective. At each step k , given Z_k , the planner selects $o_i \in \mathcal{R}_k$ so as to minimize the total number of actions α until the target is picked, i.e., $o_t \in \mathcal{P}_\alpha$, and, secondarily, the number of dropped non-target objects $|\mathcal{D}_\alpha|$. An episode fails if the target is dropped off the tabletop or if it has not been retrieved within a budget of 15 pick attempts, where an attempt is consumed whenever the planner fails to select a graspable object (after three tries), no valid grasp or motion plan is found, or a pick is executed.

4 A Modular Pipeline for TORC

Fig. 2 visualizes the modular architecture that retrieves o_t without prior object models. At each step k , the robot (1) captures Z_k and generates a segmented

point cloud, the target (if visible) labeled 0, plus a voxelization of the volume each object occludes; (2) samples grasps $g_j \in SE(3)$ per object and filters them by reachability and collision; (3) selects the next $o_i \in \mathcal{R}_k$ among the graspable objects (Sec. 5); and (4) plans and executes its retrieval with the highest-scoring grasp, after which $\mathcal{P}_{k+1} = \mathcal{P}_k \cup \{o_i\}$.

Perception. The perception module takes as input an RGB and a depth image (Fig. 2, top middle). On the real system, high-quality depth is obtained with FoundationStereo [31] from the stereo images of a ZED camera, and instance segmentation is performed on the color image via Grounding SAM 2 [15, 24], called twice: with the target’s natural-language description, and with the word “Object” to segment all objects.

An occlusion volume is maintained per object via a modified TSDF implementation [36]: occluded voxels are updated to visible ones as new snapshots are taken after each action, and each voxel stores a bitmap whose bit index is the segmentation ID of the occluding object (based on [6]).

Grasp Planning. Valid grasps tell the task planner which objects are immediately graspable, while reachable grasps in collision reveal which objects block others. Given the surface point cloud and object masks, Contact-GraspNet [27] scores candidate grasps g_j for each o_i . A candidate is valid if it admits a tabletop-collision-free, inverse-kinematics (IK) feasible solution enclosing only o_i between the fingers, with the gripper clear of the surface points of o_i (contact with its occlusion volume is permitted).

Motion Planning. Given the validated grasps for o_i , the motion planner uses CuRobo [26] for global queries and Pink⁴ for Cartesian motions; if a query fails, the next-best grasp g_{j+1} is attempted. The *approach* is a collision-free motion to a pre-grasp pose that avoids the surface points and occlusion volumes of all remaining objects, followed by a non-collision-checked Cartesian *grasp* motion to g_j , after which the gripper closes. The *extraction* lifts the object straight upward without collision checking to minimize disturbance of its neighbors; the convex hull of its point cloud and occlusion volume is then attached to the gripper as collision geometry. Finally, the *deposition* plans with this attached geometry to a drop location, avoiding the remaining objects, where o_i is released.

5 Task Planning for TORC

TORC task planning must determine a sequence of objects to pick so as to discover and retrieve the target, which may be neither visible nor pickable from the initial view, while minimizing picks and, secondarily, disturbances that cause non-target objects to fall off the tabletop. Five automated strategies are considered—RANDOM, which picks uniformly among the objects with valid grasps, a VLM approach adapted from [29], a DG approach adapted from [19], and the two proposed hybrids—plus an “expert” human-in-the-loop baseline.

⁴ <https://github.com/stephane-caron/pink>

Approach 1. VLM-SELECT.

For VLM-SELECT, a VLM is prompted with a labeled image of the workspace and asked for a sequence of actions to retrieve the target (Fig. 3).

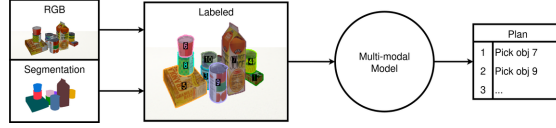


Fig. 3. VLM-SELECT: the VLM receives a labeled image and outputs the pick order.

Only the first action is executed before re-planning on a new observation. Gemini Robotics-ER 1.5 [28] is used, as it is pretrained for embodied reasoning. The prompt provides the labeled image and target ID, asks for a sequential plan that makes the target graspable (with rules about blocking and occlusion), and requests a JSON array of actions. This follows [29], minimally extended to handle initially fully occluded targets. VLM-SELECT has no knowledge of which objects have valid grasps and is re-queried (up to three times per step) if a non-graspable object is chosen.

Approach 2.

DG-SELECT. The engineered method DG-SELECT, adapted from [19], extracts dependencies from perception without object models (Fig. 4). The dependency graph $G(V, E)$ has a node $v \in V$ per visible object and a directed edge $e = (v_1 \rightarrow v_2) \in E$ indicating that v_1 cannot be picked until v_2 is picked.

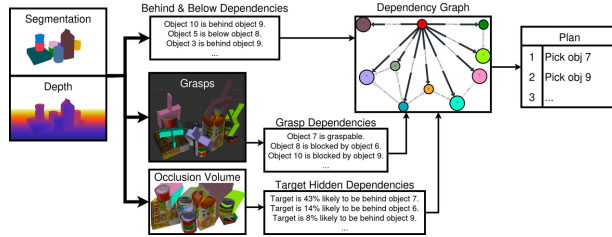


Fig. 4. DG-SELECT: “behind”/“below” relations from depth, grasp-blocking relations from the grasp planner, and target-occlusion estimates from the occlusion volumes form a dependency graph from which the next pick is selected.

To detect dependencies, each object mask is dilated and two objects are marked adjacent if their dilated masks intersect. For each adjacent pair, the average depth along their shared boundary is compared, and the object with the larger average depth is labeled “behind” the other. An adjacent pair (a, b) is instead labeled as a “below” b if a sufficient number (> 50) of boundary points on a : (1) are in close 3D proximity to b ; (2) have surface normals within 45° of the positive z axis; and (3) lie below the closest point on b .

Besides the “behind” edges H and “below” edges O , the “grasps blocked by” edges B connect each object v_1 with no valid, collision-free grasp to every object v_2 whose IK solutions for the grasps of v_1 collide with v_2 (or its occlusion volume), with edge weight $\frac{\# \text{ grasps of } v_1 \text{ blocked by } v_2}{\text{total } \# \text{ blocked grasps of } v_1}$.

$G(V, E)$ comprises all segmented objects as nodes and edges from O , B , and H added in priority order: all edges in O are added with weights $1/\text{DEG}(v_1)$; then an edge $v_1 \rightarrow v_2 \in B$ is added only if v_1 has no outgoing edge yet; the same rule then applies to H , with weights $1/\text{DEG}(v_1)$. If the target is not visible, a hidden target node t is added with a “hidden by” edge towards every segmented object, weighted proportionally to the volume of the occluded region that the

object casts. Each normalized edge weight is treated as a pseudo-probability of blocking the target: with $\mathcal{P}(t, v)$ the set of simple paths from t to a graspable sink node v , $p(v) \propto \sum_{P \in \mathcal{P}(t, v)} \prod_{e \in P} w_e$, and the next object is $v^* = \arg \max_v p(v)$.

Approach 3. VLM-FIXES-DG. The first hybrid, VLM-FIXES-DG, queries the VLM to directly output the “below” and “behind” relations between pairs of objects (Fig. 5(a)). This keeps an explicit dependency graph while reducing dependence on the hand-set depth heuristics of DG-SELECT. The prompt resembles VLM-SELECT, but the task specification also contains the JSON array of “behind” candidates derived from depth, which the VLM must correct from the image (e.g., to “below”) without adding relations, returning a JSON array of dependencies. The returned relations replace the H and O edges, while the B and hidden-target edges are computed, and the next object chosen, as in DG-SELECT.

Approach 4. VLM+GRASPS. In the second hybrid, VLM+GRASPS (Fig. 5(b)), the VLM makes the final choice of which object to pick next, but is prompted with the grasp-dependency graph, i.e., $G(V, E)$ restricted to the “grasps blocked by” edges B of DG-SELECT; it therefore uses none of the depth heuristics or their thresholds. The prompt adds to that of VLM-SELECT a scene description with sentences of the form “Object A is blocked by object B” or “Object C is graspable”. If the VLM nonetheless selects a non-pickable object, it is queried again (up to three times per step).

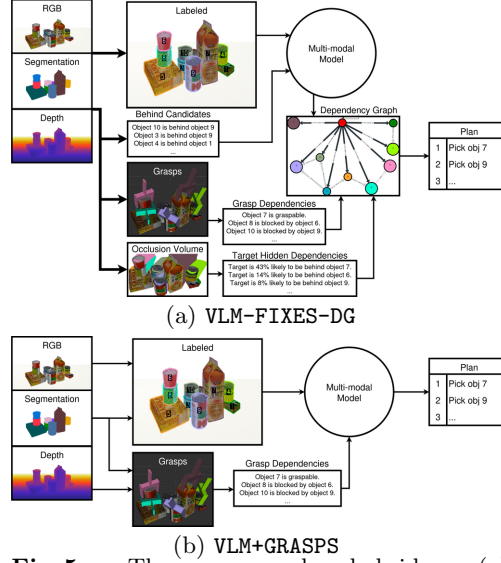


Fig. 5. The proposed hybrids. (a) VLM-FIXES-DG: the VLM outputs the “behind”/“below” relations from the labeled image and depth-derived candidates; the rest of the graph is built as in DG-SELECT. (b) VLM+GRASPS: the grasp dependencies of DG-SELECT are described in language and, with the labeled image, prompt the VLM to output the pick order.

6 Experiments

Simulated Dataset Generation. Randomly placed objects frequently yield scenes without valid grasps, so scenes were generated, labeled, tuned, and filtered to be challenging (multiple removals before the target is reachable) yet solvable, yielding 40 TORC problems (scene–target pairs) over 22 arrangements from GraspClutter6D [1], which supplies ground-truth grasps, instantiated in MuJoCo. Then: (1) an annotator labeled candidate targets, preferring occluded ones; (2) targets with a taller object in front (within 15 cm laterally) lacking

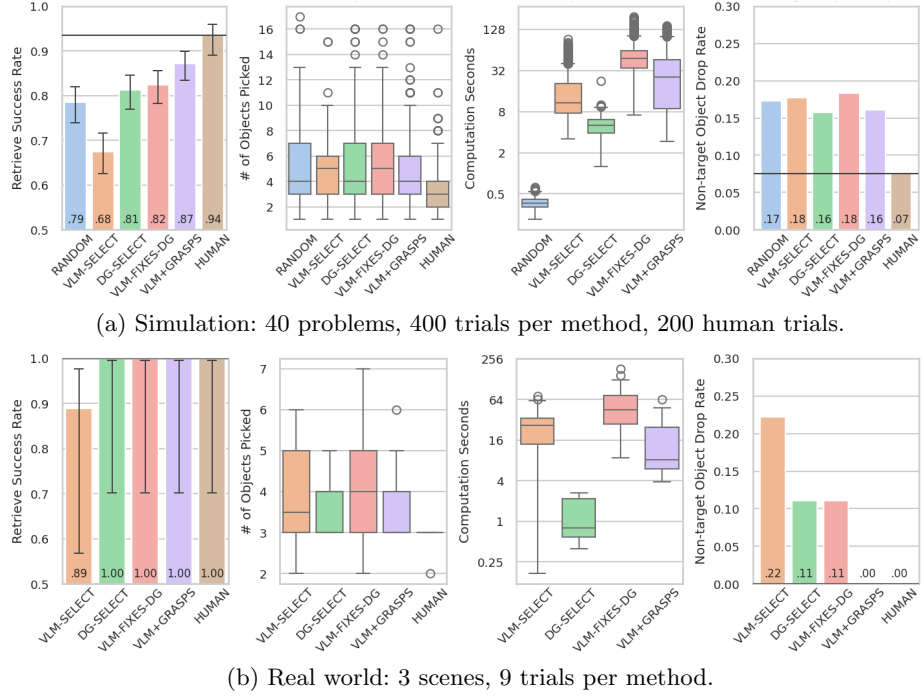


Fig. 6. Results in (a) simulation and (b) the real world. Left to right: (1) target retrieval success rate with Wilson 95% confidence intervals (axis starts at 0.5; line: human experts); (2) number of objects removed in successful trials; (3) task-planner selection time per query in seconds (log scale); (4) rate of trials in which a non-target object dropped off the tabletop (line: human experts).

a collision-free, IK-feasible ground-truth grasp were discarded; (3) scenes solvable in one pick were discarded as trivial, and unsolvable ones adjusted until an expert could solve them in at least two picks.

Simulated Evaluation. Five human experts (members of the research team) performed object selection on each of the 40 problems (200 trials), marked **HUMAN**. The experts viewed the segmented front-view image as with the VLM plus a privileged back view of the scene with the target segmented. They typed the ID of the next object to pick, without dependency or grasp information. Thus, **HUMAN** is a privileged upper bound. Fig. 6(a) reports the results.

Success rate. Using the VLM out of the box retrieves the target in only 68% of trials, *below* the 79% of **RANDOM**, which already benefits from the grasp planner since it only picks among objects with valid grasps. The engineered **DG-SELECT** reaches 81% but depends on hand-set heuristic thresholds, which were kept identical in simulation and on the real robot. **VLM-FIXES-DG**, which uses the VLM to define the “behind”/“below” relations, matches **DG-SELECT** (82%) without those depth heuristics. Describing only the threshold-free grasp dependencies to the VLM and letting it select (**VLM+GRASPS**) achieves the highest success among the automated planners, 87%, versus 94% for the human experts. Since **VLM+GRASPS**

differs from VLM-SELECT only in the language description of grasp relations, this 19-point gain isolates the value of stating geometric constraints explicitly: the model reasons well over them but cannot yet infer them reliably from the image alone. The gain bundles knowing which objects are graspable with knowing which objects block others; separating the two is left for future work.

Failure modes. Target drops end 3–5% of trials for every planner (5.5% for HUMAN), so differences stem almost entirely from budget exhaustion: 28% of VLM-SELECT trials, versus 17% for RANDOM, 14% for DG-SELECT, 12.5% for VLM-FIXES-DG, 9.5% for VLM+GRASPS, and 1% for HUMAN. In the failed VLM-SELECT trials, 8.5 of the 15 attempts on average went to objects without a valid grasp, and 72% of all its trials contained such a wasted attempt, versus 4–6% for DG-SELECT and the hybrids. RANDOM’s budget failures are instead dominated by motion-planning failures, since a valid grasp does not guarantee a feasible retrieval motion.

Scene structure. The 40 problems split evenly between *structured* arrangements (most objects in natural resting poses) and *unstructured* ones (most toppled or piled). On the 20 structured problems, VLM+GRASPS succeeds in 94% of 200 trials, statistically indistinguishable from the experts’ 90% over 100 trials (Fisher’s exact test, $p = 0.24$); on the 20 unstructured ones it drops to 81% while the experts reach 97% ($p < 10^{-4}$). A problem-level bootstrap, which respects the 10 repeated trials per problem, yields wider intervals but the same conclusions. The other DG-informed planners show no such split (80–84% on both): the VLM’s contribution is concentrated in structured scenes, and the remaining gap to humans lies in reasoning about piles.

Efficiency. In successful trials, the median number of objects removed is 3 for HUMAN, 4 for RANDOM, DG-SELECT and VLM+GRASPS, and 5 for VLM-SELECT and VLM-FIXES-DG. VLM+GRASPS thus improves success without extra picks. Selection takes 0.4 s for RANDOM and 5.1 s for DG-SELECT, whereas the VLM-based planners are bound by the remote Gemini service (10.7–47.8 s per query), VLM-FIXES-DG being the slowest.

Safety. All automated planners exhibit similar non-target drop rates (16–18% of trials), roughly twice that of the human experts (7%). Since target drops are equally frequent for humans, the non-target gap reflects selection (experts avoid picks that destabilize neighbors), whereas the drops themselves arise in execution: without object models, the picked object’s geometry is approximated by the convex hull of its visible points and occlusion volume, and objects leaning on or stacked over it may lose their support. Objects that settle off the table when an arrangement is instantiated in MuJoCo (a minority of problems) are never targets and are excluded from this count.

Real-World Evaluation. The real-world experiments used a single arm of a Yaskawa Motoman SDA10F robot with a Robotiq 85 gripper and a ZED Mini stereo camera (Fig. 1). Three scenes of household objects were built on a grid for replication. Each scene required at least two picks and avoided segmentation and grasp-planner failures, isolating object selection. Each of the four VLM/DG planners and the human experts were evaluated three times per scene (9 trials



Fig. 7. Two real-world retrieval sequences (rows): the segmented robot view before each of three picks. The target (label 0, red silhouette) is initially hidden and becomes graspable only after two removals.

each, 45 in total); **RANDOM** was omitted to limit robot time, since its selection does not depend on the image modality that separates real from simulated input. Fig. 6(b) reports the results and Fig. 7 shows example sequences.

Real-world observations mirror simulation. Still, **VLM-SELECT** was the only automated method to fail (89% success; in the failed trial, 9 of the 15 attempts were spent selecting non-graspable objects). It also had the highest drop rate (22%), while the **DG**-based selectors (**DG-SELECT**, **VLM-FIXES-DG**) retrieved the target in every trial but each dropped a non-target object once. **VLM+GRASPS** solved all 9 trials without dropping any object and with a median of 3 picks, matching the human experts.

7 Discussion and Conclusion

This work evaluates **VLMs** as task planners for **Targeted Object Retrieval in Clutter (TORC)**, comparing and integrating them with geometric, graph-based solutions in a complete pipeline for unknown objects. Three observations emerge. First, spatial reasoning from a single image is not yet sufficient for a pretrained **VLM** to sequence picks in dense clutter. Without explicit reachability information, it does not reliably identify which objects block the target. Second, the missing geometric information is cheap to compute from the grasp planner. Once expressed in language, the **VLM** uses it effectively, outperforming the fully engineered **DG** pipeline without any of its thresholds, and on structured scenes it is statistically on par with experts who had a privileged view; the remaining gap is in unstructured piles. Third, letting the **VLM** decide given explicit constraints (**VLM+GRASPS**) beats letting it populate the graph (**VLM-FIXES-DG**). This implies that, at least for the model studied here, **VLMs** currently serve better as reasoners over structured evidence than as relation detectors.

The study has limitations: 40 curated simulated problems and three real scenes, all from a single front-facing viewpoint; hand-set **DG** thresholds; ground-truth segmentation in simulation; and a single **VLM**. The drop rate of every automated planner also stays above that of humans, given the lack of object models and the convex-hull reconstruction of the picked object. Finally, the hybrids can generate manipulation sequences for **TORC** as data for fine-tuning future **VLMs** and vision-language-action models, enabling more reactive, closed-loop retrieval.

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